

Ritz Method — Gelfand - Fomin Problem 8.2

Austin Shank

2026-08-14

1 Problem Statement

2 Solution

2.1 Exact Solution

We will define the julia function of this for comparison against the approximations

```
y_exact(x) = sin(x) / sin(1) - x  
have_exact = !isnothing(y_exact)
```

2.2 Ritz Approximations

We will use the RitzMethod package

```
using RitzMethod  
import Optim  
import Plots  
  
# Define problem and basis  
integrand(x, y, y_prime) = y_prime^2 - y^2 - 2 * x * y  
domain = Interval(0.0, 1.0)  
full_basis = [x -> x^k * (1 - x) for k in 1:3]  
  
function solve_ritz(n_terms)  
    minimizefunctional(  
        integrand, domain, full_basis[1:n_terms];  
    )  
end
```

2. Use the Ritz method to find an approximate solution of the problem of minimizing the functional

$$J[y] = \int_0^1 (y'^2 - y^2 - 2xy) dx, \quad y(0) = y(1) = 0,$$

and compare the answer with the exact solution.

Figure 1: Problem 8.2

8.2	8/15/2026	
Problem: Find the extremal of $J[y] = \int_0^1 y'^2 - y^2 - 2xy \, dx$, $y(0) = y(1) = 0$ Soln: $F(x, y, y') = y'^2 - y^2 - 2xy$ $\left. \begin{aligned} F_y &= -2y - 2x \\ F_{y'} &= 2y' \\ \frac{d}{dx} F_{y'} &= 2y'' \end{aligned} \right\} \Rightarrow -2y - 2x - 2y'' = 0 \Rightarrow y'' + y = -x$ Solve Euler Eqn homogeneous: $y'' + y = 0 \Rightarrow m^2 + 1 = 0 \Rightarrow m = \pm i \Rightarrow y_h = C_1 \cos x + C_2 \sin x$ particular soln: $y_p = Ax + B, \quad y_p' = A, \quad y_p'' = 0$ substitute $Ax + B = -x \Rightarrow A = -1, B = 0 \Rightarrow y_p = -x$ general soln $y_g = C_1 \cos x + C_2 \sin x - x$ Solve for C_1, C_2 $\left. \begin{aligned} y(0) = 0: & 0 = C_1 \\ y(1) = 0: & 0 = C_2 \sin 1 - 1 \Rightarrow C_2 = \frac{1}{\sin 1} \end{aligned} \right\} \Rightarrow \boxed{y_g = \frac{\sin x}{\sin 1} - x}$		

Figure 2: Problem 8.2

```

        num_quad_nodes=5,
        method=Optim.BFGS(),
        options=Optim.Options(iterations=50, g_abstol=1e-20, x_reltol=1e-8),
    )
end

results = [solve_ritz(n) for n in 1:3]

```

2.2.1 One-Term Approximation

```
println(summarize(results[1]))
```

RitzResult:

```

Domain:          [0.0, 1.0]
Coefficients:    [0.2777777777777778]
Minimized value: -0.02314814814814817
Converged:       true

```

2.2.2 Two-Term Approximation

```
println(summarize(results[2]))
```

RitzResult:

```

Domain:          [0.0, 1.0]
Coefficients:    [0.19241192411920088, 0.1707317073170797]
Minimized value: -0.024570912375790444
Converged:       true

```

2.2.3 Three-Term Approximation

```
println(summarize(results[3]))
```

RitzResult:

```

Domain:          [0.0, 1.0]
Coefficients:    [0.18776681077849913, 0.1941430785544192, -0.02341137123
737456]
Minimized value: -0.024574009118017617
Converged:       true

```

3 Comparison

Term count	Minimized J[y]	Coefficients
1	-0.02314815	[0.2777777777777778]
2	-0.02457091	[0.19241192411920088, 0.1707317073170797]
3	-0.02457401	[0.18776681077849913, 0.1941430785544192, -0.02341137123737456]

4 Approximation Error

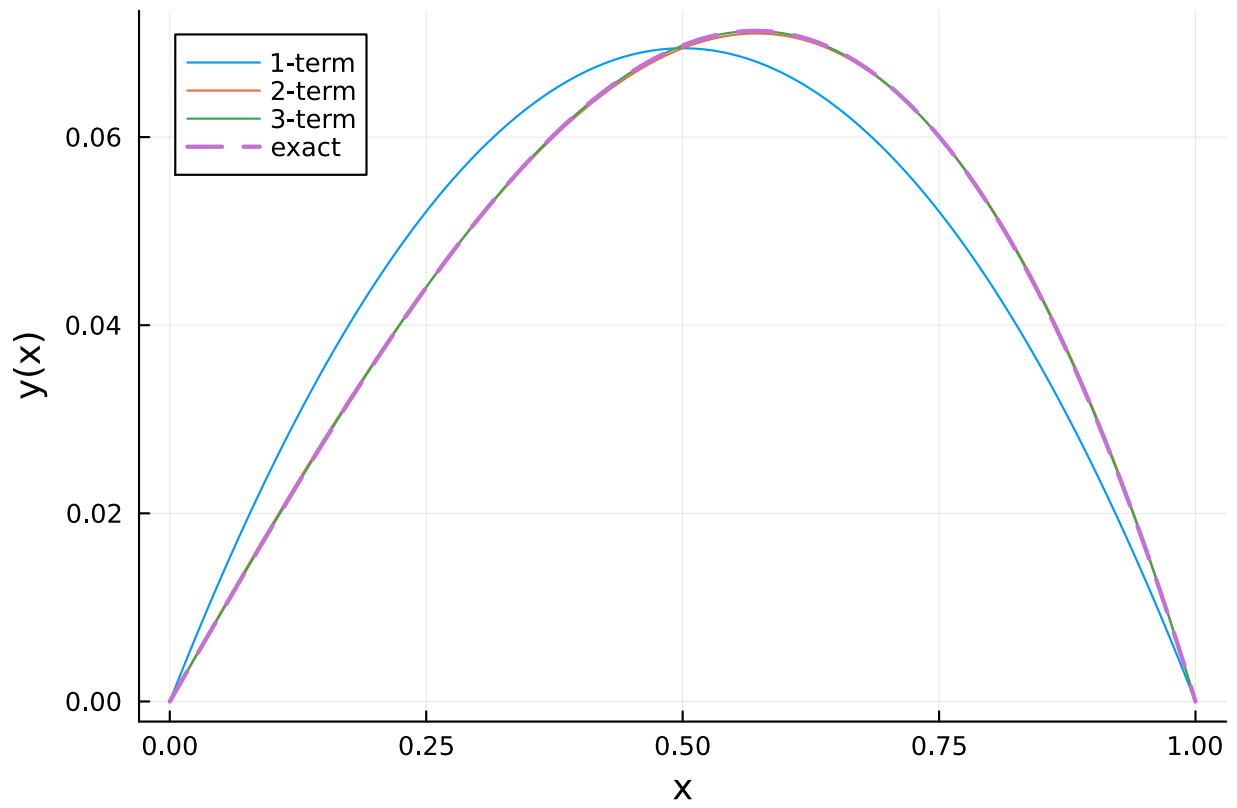


Figure 3: Ritz Approximations

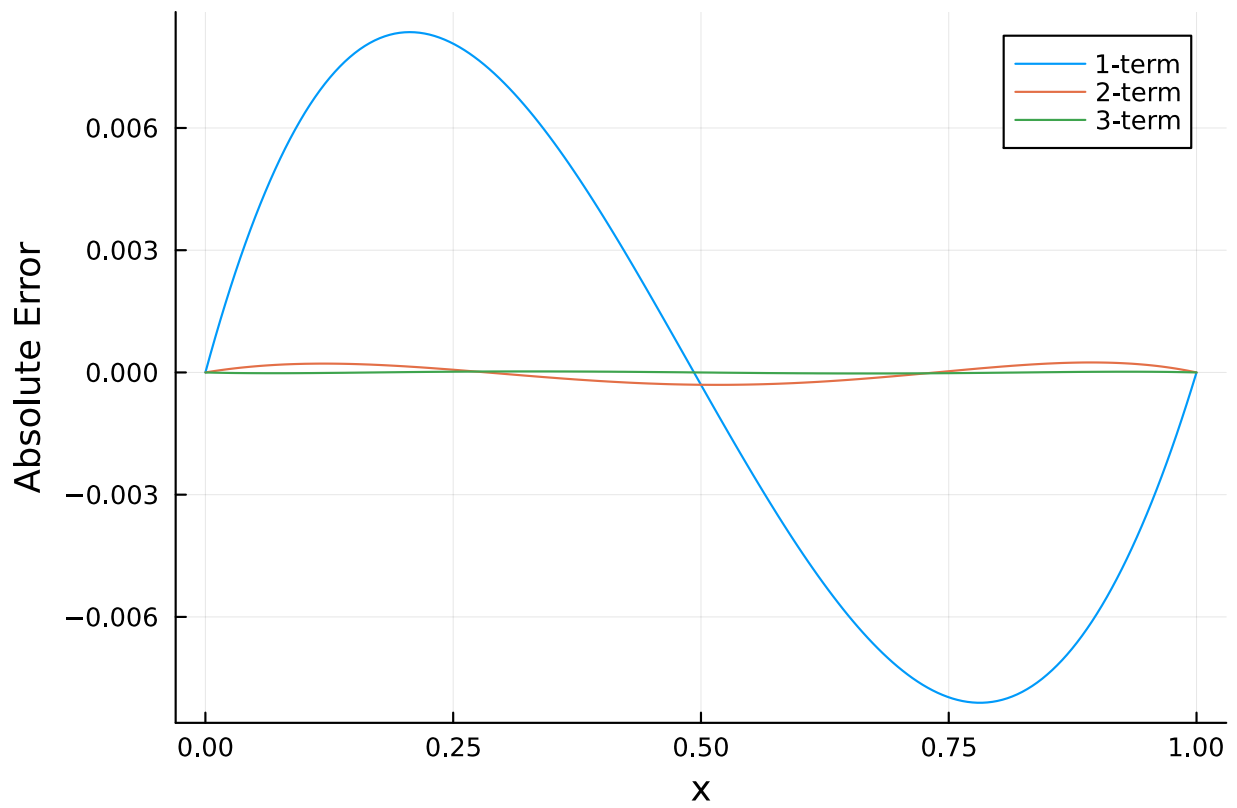


Figure 4: Pointwise error between Ritz approximations and the exact solution.